

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 08: JOURNEY INSIDE THE ATOM: STRUCTURE & COMPOSITION
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-08

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): Dalton's atomic theory postulate that atoms are indivisible particles was disproved by late 19th-century physics discoveries.

Reason (R): The experimental discovery of subatomic particles (electrons, protons, and neutrons) proved that atoms are composed of smaller constituent entities.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): Cathode rays discovered by J.J. Thomson consist of streams of negatively charged particles called electrons.

Reason (R): Cathode rays are deflected towards the positive plate in an external electric field and possess an identical charge-to-mass ratio (e/m_e) regardless of the gas used in the discharge tube.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): An atom as a whole is electrically neutral.

Reason (R): The total number of negatively charged electrons orbiting outside the nucleus equals the total number of positively charged protons located inside the nucleus.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): J.J. Thomson proposed the 'plum pudding' model of the atom.

Reason (R): Thomson visualized the atom as a uniform positively charged sphere in which electrons are embedded like dry raisins in a pudding.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): In Rutherford's alpha particle scattering experiment, a thin foil of gold was selected as the target.

Reason (R): Gold is the most malleable metal, allowing fabrication of extremely thin foils approximately 1,000 atoms thick.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): Most of the alpha particles passed straight through the gold foil without any deflection.**Reason (R):** Most of the space inside an atom is completely empty.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.7

Assertion (A): Rutherford deduced that the entire positive charge and nearly all the mass of an atom are concentrated in a tiny central region called the nucleus.**Reason (R):** A small fraction of alpha particles (about 1 in 12,000) suffered massive deflections and rebounded backward at 180°.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.8

Assertion (A): Rutherford's planetary model of the atom could not explain the atomic stability of matter.**Reason (R):** According to classical electromagnetic theory, an accelerating charged electron orbiting a nucleus should continuously radiate electromagnetic energy, lose speed, and spiral into the nucleus in 10^{-8} s.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.9

Assertion (A): Neils Bohr proposed that electrons revolve around the nucleus only in certain discrete, non-radiating orbits called stationary energy levels.**Reason (R):** While revolving in these discrete permissible orbits, electrons do not radiate energy.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.10

Assertion (A): The maximum number of electrons that can be accommodated in the n^{th} -th energy shell is given by $2n^2$.

Reason (R): For the M-shell ($n=3$), the maximum theoretical capacity according to the Bohr-Bury scheme is 18 electrons.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): The outermost valence shell of an atom cannot accommodate more than 8 electrons, even if its theoretical capacity is larger.

Reason (R): An octet of valence electrons confers exceptional chemical stability (octet rule), as observed in noble gas electronic configurations.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): James Chadwick discovered the neutron by bombarding thin beryllium targets with alpha particles in 1932.

Reason (R): Neutrons are electrically neutral subatomic particles with a mass approximately equal to that of a proton, located within atomic nuclei.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): The mass number (A) of an atom is defined as the total number of protons and neutrons (nucleons) present in its nucleus.

Reason (R): Electrons have negligible mass ($\approx 1/1836$ of a proton mass), so the entire mass of an atom is concentrated in its nucleus.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Valency of magnesium (atomic number 12) is 2, while valency of chlorine (atomic number 17) is 1.

Reason (R): Magnesium readily loses 2 valence electrons ($2, 8, 2$) to achieve an octet, while chlorine gains 1 electron ($2, 8, 7$) to complete its octet ($8 - 7 = 1$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): Hydrogen is the only element whose most abundant atom (${}^1_1\text{H}$, protium) contains no neutrons in its nucleus.

Reason (R): Protium has an atomic number of 1 and mass number of 1, yielding $A - Z = 1 - 1 = 0$ neutrons.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): Isotopes of an element exhibit identical chemical properties.

Reason (R): Isotopes have the exact same atomic number and electronic configuration, and chemical reactivity is determined exclusively by valence electrons.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): Isotopes of an element exhibit slightly different physical properties such as density and boiling points.

Reason (R): Isotopes possess different numbers of neutrons and consequently different atomic masses.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): The average atomic mass of chlorine is recorded as 35.5 u.

Reason (R): Natural chlorine occurs as a mixture of two isotopes, ^{35}Cl and ^{37}Cl , in a relative abundance ratio of 3:1 (75% to 25%).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): Cobalt-60 (^{60}Co) isotope is widely used in the medical treatment of cancer (radiotherapy).

Reason (R): Cobalt-60 emits high-energy penetrating gamma rays that destroy rapidly dividing malignant tumor cells.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): Uranium-235 (^{235}U) isotope is utilized as fuel in nuclear fission reactors to generate electrical power.

Reason (R): The fission of Uranium-235 nuclei induced by thermal neutrons releases enormous quantities of controlled nuclear energy.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): Iodine-131 (^{131}I) isotope is used in the clinical diagnosis and treatment of goitre and thyroid disorders.

Reason (R): The thyroid gland selectively absorbs and concentrates iodine from the bloodstream to synthesize thyroxine hormone.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): $^{40}_{18}\text{Ar}$ (Argon) and $^{40}_{20}\text{Ca}$ (Calcium) are classified as isobars.

Reason (R): Isobars are atoms of different chemical elements having different atomic numbers but identical mass numbers ($A=40$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): An alpha particle is essentially a doubly charged helium ion (He^{2+}).

Reason (R): An alpha particle consists of two protons and two neutrons, with mass 4 u and charge $+2e$, identical to a helium nucleus stripped of both electrons.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): Noble gases like helium, neon, and argon have zero chemical valency.

Reason (R): Noble gases possess completely filled valence electron shells (duplet in helium, octets in neon and argon), rendering them chemically inert under standard conditions.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): The radius of the atomic nucleus is approximately 10^5 times smaller than the radius of the whole atom.

Reason (R): Nuclear radius is on the order of 10^{-15} m (fermi), whereas atomic radius is on the order of 10^{-10} m (angstrom).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and explains how cathode ray and radioactivity experiments proved atoms are divisible.
Q.2	(A)	Assertion is true. Reason is true and proves electrons are universal subatomic constituents of all matter.
Q.3	(A)	Assertion is true. Reason is true and explains electrostatic balance in neutral atoms ($+Ze + (-Ze) = 0$).
Q.4	(A)	Assertion is true. Reason is true and describes Thomson's classical atomic model.
Q.5	(A)	Assertion is true. Reason is true and justifies why gold was chosen to minimize multiple scatterings.
Q.6	(A)	Assertion is true. Reason is true and provides Rutherford's primary deduction from undeflected alpha trajectories.
Q.7	(A)	Assertion is true. Reason is true and explains the experimental evidence for a dense, massive, positive nucleus.
Q.8	(A)	Assertion is true. Reason is true and states the fundamental theoretical collapse of Rutherford's classical model.
Q.9	(A)	Assertion is true. Reason is true and gives Bohr's postulate resolving atomic stability.
Q.10	(B)	Both statements are true ($2(3)^2 = 18$), but Reason calculates the value for M-shell rather than deriving the $2n^2$ quantum rule.
Q.11	(A)	Assertion is true. Reason is true and explains the chemical stability associated with an 8-electron valence shell.
Q.12	(B)	Both statements are historically and physically true, but Reason describes neutron properties rather than explaining the nuclear reaction (${}^9_4\text{Be} + {}^4_2\alpha \rightarrow {}^{12}_6\text{C} + {}^1_0\text{n}$).
Q.13	(A)	Assertion is true. Reason is true and explains why mass number ignores orbiting electrons.
Q.14	(A)	Assertion is true. Reason is true and explains valency in terms of electron loss and gain to attain noble gas octet.
Q.15	(A)	Assertion is true. Reason is true and shows mathematical proof ($N = A - Z = 0$) for protium.
Q.16	(A)	Assertion is true. Reason is true and explains why identical valence electron structure ensures identical chemical behavior.
Q.17	(A)	Assertion is true. Reason is true and explains how mass differences affect physical bulk properties.
Q.18	(A)	Assertion is true. Reason is true and correctly calculates: $\frac{(35 \times 3) + (37 \times 1)}{4} = \frac{142}{4} = 35.5 \text{ u}$.
Q.19	(A)	Assertion is true. Reason is true and provides the radiobiological mechanism of Cobalt-60 cancer therapy.
Q.20	(A)	Assertion is true. Reason is true and explains how U-235 sustains controlled chain fission reactions.
Q.21	(A)	Assertion is true. Reason is true and explains the organ-specific uptake of iodine by the thyroid gland.
Q.22	(A)	Assertion is true. Reason is true and defines isobars with matching mass numbers.
Q.23	(A)	Assertion is true. Reason is true and provides the nuclear composition of alpha radiation.
Q.24	(A)	Assertion is true. Reason is true and explains why full valence shells lead to zero valency.
Q.25	(A)	Assertion is true. Reason is true and mathematically confirms the dimensional ratio: $10^{-10} / 10^{-15} = 10^5$.